

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 03
Duration: 20 Minutes

Semester : Fall25
Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:09
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1. Consider the following SDT. Draw the Annotated Parse Tree and Evaluate the values of the attributes type and width along with variable offset next to the appropriate non-terminal nodes for the following String:

```
record {
int a;
record {
float b;
}r1;
}recor;
float [5] [7] z;
```

Afterwards draw the memory diagram along with actual location and data

Note: Offsets are given in decimal. Memory location is also given in decimal. Every memory unit is of 1 byte. The width of int and float is also given in byte in the SDT. The variable memory_addr refers to the starting position of the memory address where the variables will

$P \rightarrow \{ \text{offset} = 0; \text{memory_addr} = 58342; \} D$

$D \rightarrow T \text{ id } ; \{ \text{top.put}(\text{id.lexeme}, T.\text{type}, \text{offset}); \quad D_1$
 $\text{offset} = \text{offset} + T.\text{width}; \}$

$D \rightarrow \epsilon$

$T \rightarrow \text{record '}' \{ \text{Env.push}(\text{top}); \text{top} = \text{new Env}(); \quad D \text{ '}' \} \quad \{ T.\text{type} = \text{record}(\text{top}); T.\text{width} = \text{offset};$
 $\text{Stack.push}(\text{offset}); \text{offset} = 0; \}$ $\text{top} = \text{Env.pop}(); \text{offset} = \text{Stack.pop}();$

$T \rightarrow B \{ t = B.\text{type}; w = B.\text{width}; \} C$

$\{ T.\text{type} = C.\text{type}; T.\text{width} = C.\text{width}; \}$

$B \rightarrow \text{int}$

$\{ B.\text{type} = \text{integer}; B.\text{width} = 3; \}$

$B \rightarrow \text{float}$

$\{ B.\text{type} = \text{float}; B.\text{width} = 7; \}$

$C \rightarrow \epsilon$

$\{ C.\text{type} = t; C.\text{width} = w; \}$

$C \rightarrow [\text{num}] C_1$

$\{ C.\text{type} = \text{array}(\text{num.value}, C_1.\text{type});$
 $C.\text{width} = \text{num.value} \times C_1.\text{width}; \}$

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